

UNIT -1

Introduction

SQL (Structured Query Language) is used to create, manage and retrieve data from databases. A database organizes data using database software, databases, schemas, tables and columns. It helps users create databases and tables and work with the stored data. MySQL is a commonly used database software.

Database Structure

- Database Software: Software such as MySQL, PostgreSQL and MS SQL.
- Database: Stores data entities with different purposes.
- Schema: A collection of related tables.
- Table: Stores data in rows and columns.
- Data: The actual information stored in the database.

Basic SQL Commands

Command	Use
CREATE	Schema/table
SHOW	View information
DROP	Delete table
TRUNCATE	Delete table data
ALTER	Modify table

Data Types

- Number: INT, FLOAT, DECIMAL
- Date: DATE, DATETIME
- Text: CHAR, VARCHAR, TEXT

Column Attributes

- NOT NULL → Field must contain a value.
- AUTO_INCREMENT → Automatically generates numbers.
- DEFAULT → Provides a default value.

These help maintain correct and consistent data in table

Analytical Reporting Case Study

#Create database and use the database

```
create database movies_db;  
use movies_db;
```

```
create table movies2  
(  
    rel_year int,  
    mov_name varchar(100),  
    act_name varchar(50),  
    lang varchar(30),  
    dir_name varchar(100)  
);
```

#Insert the data's

```
INSERT INTO MOVIES2 VALUES (1963,'Anandha Jodhi','Kamal Haasan','Tamil','V. N. Reddy');  
INSERT INTO MOVIES2 VALUES (1963,'Vanambadi','Kamal Haasan','Tamil','G. R. Nathan');  
INSERT INTO MOVIES2 VALUES (1970,'Maanavan','Kamal Haasan','Tamil','M. A. Thirumugam');  
INSERT INTO MOVIES2 VALUES (1971,'Annai Velankanni','Kamal Haasan','Tamil','Thankappan');  
INSERT INTO MOVIES2 VALUES (1971,'Anubhavangal Paalichakal','Mammootty','Malayalam','K. S. Sethumadhavan');  
INSERT INTO MOVIES2 VALUES (1972,'Kurathi Magan','Kamal Haasan','Tamil','K. S. Gopalakrishnan');  
INSERT INTO MOVIES2 VALUES (1973,'Arangetram','Kamal Haasan','Tamil','K. Balachander');  
INSERT INTO MOVIES2 VALUES (1973,'Kaalachakram','Mammootty','Malayalam','K. Narayanan');  
INSERT INTO MOVIES2 VALUES (1973,'Sollathaana Ninaikkiren','Kamal Haasan','Tamil','K. Balachander');  
INSERT INTO MOVIES2 VALUES (1974,'Anbu Thangai','Kamal Haasan','Tamil','S. P. Muthuraman');  
INSERT INTO MOVIES2 VALUES (1974,'Aval Oru Thodar Kathai','Kamal Haasan','Tamil','K. Balachander');  
INSERT INTO MOVIES2 VALUES (1974,'Gumasthavin Magal','Kamal Haasan','Tamil','A. P. Nagarajan');  
INSERT INTO MOVIES2 VALUES (1974,'Kanyakumari','Kamal Haasan','Malayalam','K. S. Sethumadhavan');
```

1.Retrieve all details from the movie table.

```
SELECT * FROM MOVIES2;
```

Solution:

	rel_year	mov_name	act_name	lang	dir_name
►	1960	Kalathur Kannamma	Kamal Haasan	Tamil	A. Bhimsingh
	1962	Kannum Karalum	Kamal Haasan	Malayalam	K. S. Sethumadhavan
	1962	Paadha Kaanikkai	Kamal Haasan	Tamil	K. Shankar
	1962	Paarthaal Pasi Theerum	Kamal Haasan	Tamil	A. Bhimsingh
	1963	Anandha Jodhi	Kamal Haasan	Tamil	V. N. Reddy
	1963	Vanambadi	Kamal Haasan	Tamil	G. R. Nathan
	1970	Maanavan	Kamal Haasan	Tamil	M. A. Thirumugam
	1971	Annai Velankanni	Kamal Haasan	Tamil	Thankappan
	1971	Anubhavangal Paalichakal	Mammootty	Malayalam	K. S. Sethumadhavan
	1972	Kurathi Magan	Kamal Haasan	Tamil	K. S. Gopalakrishnan
	1973	Arangetram	Kamal Haasan	Tamil	K. Balachander
	1973	Kaalachakram	Mammootty	Malayalam	K. Narayanan
	1973	Sollathaana Ninaikkiren	Kamal Haasan	Tamil	K. Balachander
	1974	Anbu Thangai	Kamal Haasan	Tamil	S. P. Muthuraman
	1974	Aval Oru Thodar Kathai	Kamal Haasan	Tamil	K. Balachander
	1974	Gumasthavin Magal	Kamal Haasan	Tamil	A. P. Nagarajan
	1974	Kanyakumari	Kamal Haasan	Malayalam	K. S. Sethumadhavan
	1974	Naan Avanillai	Kamal Haasan	Tamil	K. Balachander
	1974	Panathukkaga	Kamal Haasan	Tamil	M. S. Senthil

2.Display only movie names and actor names.

```
SELECT mov_name, act_name FROM MOVIES2;
```

Solution:

mov_name	act_name
Kalathur Kannamma	Kamal Haasan
Kannum Karalum	Kamal Haasan
Paadha Kaanikkai	Kamal Haasan
Paarthaal Pasi Theerum	Kamal Haasan
Anandha Jodhi	Kamal Haasan
Vanambadi	Kamal Haasan
Maanavan	Kamal Haasan
Annai Velankanni	Kamal Haasan
Anubhavgal Paalichakal	Mammootty
Kurathi Magan	Kamal Haasan
Arangetram	Kamal Haasan
Kaalachakram	Mammootty
Sollathaen Ninaikkiren	Kamal Haasan
Anbu Thangai	Kamal Haasan
Aval Oru Thodar Kathai	Kamal Haasan
Gumasthavin Magal	Kamal Haasan
Kanyakumari	Kamal Haasan
Naan Avanillai	Kamal Haasan
Paruthi Idavara	Kamal Haasan

3.Display distinct movie languages.

```
SELECT DISTINCT lang FROM MOVIES2;
```

Solution:

lang
Tamil
Malayalam

4.Display distinct movie languages.

```
SELECT DISTINCT act_name FROM MOVIES2;
```

Solution:

act_name
Kamal Haasan
Mammootty
Rajinikanth

Conclusion

SQL Data Structure provides a basic understanding of databases, schemas, tables, data types and column attributes. In conclusion, SQL Data Structure helps us understand how data is organized and managed in a database. It includes database software, databases, schemas, tables, data types, and column attributes. SQL commands help us create, view, modify, and manage tables and data efficiently.